Store-Level Restaurant Visitor Forecasting: A Seasonal Time Series Comparison

Evidence from the Recruit Restaurant Visitor Forecasting Dataset

Contents

Scope statement. This report presents only the methods and results implemented in the final project pipeline. It does not rank models outside the final store-level notebook. The central claim is deliberately narrow: on the main 250-store evaluation sample, a holiday-only SARIMAX specification is compared with Seasonal Naive lag 7 and SARIMA no exog using pooled store-date RMSLE on the final chronological holdout and three rolling-origin validation folds.

Abstract

This report studies daily restaurant visitor forecasting using the Recruit Restaurant Visitor Forecasting dataset. The forecasting unit is `air_store_id` × `visit_date`, and the empirical comparison is intentionally layered: a weekly Seasonal Naive lag-7 benchmark, a store-level SARIMA model with no exogenous regressors, and a compact holiday-only SARIMAX extension. The design is meant to identify where predictive gains come from in a disciplined time-series setting rather than to run a broad model tournament.

Restaurant demand is treated as an operational decision problem under uncertainty. Daily visitors are shaped by weekly repetition, store-specific stochastic dependence, and known calendar shocks. The final pipeline builds store-level daily series, applies a `log1p` transformation to visitors, and estimates fixed-order store-specific models with seasonal period 7. The main evaluation sample contains 250 eligible AIR stores. The final holdout runs from 15 March 2017 to 22 April 2017 and contains 9,721 observed store-date rows.

On that holdout, pooled RMSLE is 0.9314 for Seasonal Naive, 0.7255 for SARIMA no exog, and 0.7098 for holiday-only SARIMAX. Relative to Seasonal Naive, SARIMA reduces RMSLE by 22.1%, while holiday-only SARIMAX reduces RMSLE by 23.8%. Relative to SARIMA, the holiday-only specification improves RMSLE by a further 2.17% and outperforms SARIMA for 67.2% of stores. Three rolling-origin folds provide supporting temporal evidence: holiday-only SARIMAX is best in two folds, while SARIMA is best in one. The overall conclusion is clear but restrained: most predictive gain comes from modelling weekly seasonal dynamics correctly, while holiday information provides a modest additional improvement.

1. Introduction

Restaurant managers must make daily staffing, inventory, and capacity decisions before realised demand is fully observed. If the forecast is too low, the restaurant may face long waiting times, overworked staff, stock-outs, and lower customer satisfaction. If the forecast is too high, the restaurant may waste labour hours and perishable ingredients. Visitor forecasting is therefore not merely a technical prediction exercise. It is an operational problem of allocating limited resources under uncertainty, where forecast quality matters because it shapes decisions taken before demand is known.

This report addresses a deliberately narrow question: once a strong weekly benchmark is established, does a compact exogenous extension still add value? More precisely, the project asks whether a store-level holiday-only SARIMAX specification can improve forecast accuracy relative to both a simple weekly Seasonal Naive benchmark and a store-level SARIMA model with no exogenous regressors. The scope is intentionally precise. The project does not attempt to rank every possible forecasting family. It instead evaluates a layered time-series question motivated by the economics of restaurant demand: how much predictive value comes from repeated weekly structure, how much comes from stochastic seasonal dynamics, and whether a known holiday signal adds further information beyond those internal dynamics.

The empirical setting is the Recruit Restaurant Visitor Forecasting dataset, which contains visitor counts, reservation records, store information, and calendar variables for restaurants in Japan. The final modelling notebook begins with 814 AIR stores, identifies 812 stores that satisfy the modelling requirements, and then evaluates a main sample of 250 eligible stores chosen for the final comparison. This 250-store sample is the report's main estimation-and-evaluation sample rather than a casual illustration: it is a fixed subset of eligible stores used consistently across the holdout comparison and the three rolling-origin folds so that runtime, model comparability, and evidentiary traceability remain aligned. The final chronological holdout runs from 15 March 2017 to 22 April 2017. The unit of analysis is a store-date row. The project is therefore not forecasting one aggregate market-level series; it is forecasting daily demand for each individual store, which matches the real decision problem faced by restaurant operators and platforms.

Within that scope, the report makes four contributions. First, it aligns the empirical design with the operational task by using store-level daily forecasting rather than aggregate forecasting. Second, it structures the comparison as a transparent ladder: Seasonal Naive lag 7, SARIMA no exog, and holiday-only SARIMAX. Third, it treats pooled store-date RMSLE as the primary accuracy metric while using store-level win rates as supporting evidence that any gain is broadly distributed rather than concentrated in a few restaurants. Fourth, it validates the ranking with three rolling-origin folds instead of relying on a single chronological split.

The main result is straightforward. Most of the predictive gain comes from moving from Seasonal Naive to SARIMA, which indicates that modelling seasonal stochastic dependence is already doing most of the work. The holiday-only SARIMAX specification then improves modestly further on the final holdout and in two of the three rolling-origin folds. The supported conclusion is therefore specific: weekly repetition matters, seasonal time-series structure matters more, and a parsimonious holiday regressor adds a small but interpretable incremental gain.

The rest of the report is organised as follows. Section 2 reviews the relevant forecasting literature and develops the theoretical framework. Section 3 describes the data, sample, variables, models, and evaluation strategy. Section 4 presents the main results. Section 5 discusses the economic interpretation, limitations, and future research directions. Section 6 concludes.

1.1 Store-level forecasting as the decision unit

The store-level design is central to the logic of the project. A platform may care about total demand, but staffing, food preparation, and service bottlenecks are all store-specific. An aggregate forecast can therefore look accurate while still being operationally useless if it hides large errors across individual restaurants.

This point also strengthens the use of pooled store-date RMSLE. The forecast task is not one series and one error per day; it is a panel of store-date forecasts. Pooled RMSLE treats each evaluated store-date row as an operational prediction unit and summarises performance across that panel. The store-level win rate then adds a second diagnostic by asking whether a model improves broadly across stores or whether the pooled result is driven by a small subset of restaurants.

The store-level design does introduce a trade-off. Estimating separate SARIMA- or SARIMAX-family models by store means each model has fewer observations than a pooled global model would have. Some stores have shorter histories and noisier demand patterns. However, this trade-off is reasonable for the goal of the report. Separate store-level models allow each restaurant's history to drive its own forecast, which is consistent with the theoretical view that restaurants differ in clientele, location, cuisine, and operating schedule.

2. Literature Review and Theoretical Framework

2.1 What the literature suggests about restaurant demand forecasting

The literature relevant to this project spans three connected questions rather than one isolated modelling choice. First, how should restaurant demand be conceptualised as a time series? Second, what counts as a fair benchmark in a forecasting study? Third, when does external information justify moving from a pure univariate model to a dynamic regression design? Organising the review around these questions makes the empirical logic of the project clearer than simply listing forecasting methods in chronological order.

The first strand is the classical time-series literature. Box and Jenkins (1970) provide the core logic of ARIMA modelling, in which current values depend on autoregressive structure, differencing, and moving-average errors. For seasonal data, this framework naturally extends to seasonal autoregressive and seasonal moving-average components. Restaurant visitor counts are an intuitively suitable application because they are shaped by repeated weekly routines: Monday demand differs from Friday demand, and weekends often follow a different rhythm from workdays. The key implication is not merely that ARIMA exists, but that restaurant demand is unlikely to be well described as iid noise. It is a structured stochastic process with persistence and seasonality.

The second strand concerns forecasting benchmarks. Hyndman and Athanasopoulos (2021) argue that more complex models should be judged against simple and transparent alternatives, not against weak strawman comparators. This point is reinforced by large-scale forecasting evidence. Makridakis, Spiliotis, and Assimakopoulos (2018) show that simple statistical benchmarks remain difficult to beat across many forecasting environments. In the context of restaurant demand, this makes a weekly Seasonal Naive lag-7 forecast the relevant benchmark rather than a mean forecast or an unconditional trend rule. If demand has strong weekly structure, then repeating the same weekday from the previous week is already a serious forecasting standard. The empirical question is therefore not whether a structured model can outperform something trivial, but whether it can add value beyond a benchmark that already captures the most obvious seasonal pattern.

The third strand concerns dynamic regression and forecast-origin-valid exogenous information. In seasonal settings, the more precise label is SARIMAX: a seasonal ARIMA structure augmented with predetermined external regressors. For restaurant demand, public holidays are natural candidates for this role because they alter leisure patterns, timing of outings, and local demand rhythms in ways that may not be fully inferred from recent weekly history alone. The intervention-analysis tradition of Box and Tiao (1975) is relevant here because it formalises the idea that known external events should not simply be left in the error term when they are observable in advance.

A related methodological lesson concerns forecast evaluation. Hyndman and Koehler (2006) show that commonly used percentage-error measures can behave poorly when realised values are close to zero.

That warning matters in store-level restaurant data, where some store-days have low visitor counts. The implication is broader than metric selection alone: evaluation should match the structure of the operational problem. If the decision unit is a store-date forecast, then the primary error metric should also respect that unit and avoid exaggerating the influence of low-count denominators.

Taken together, this literature creates a useful scholarly tension. On one side, simple weekly benchmarks are often strong and difficult to beat. On the other side, restaurant demand is not generated purely by literal last-week repetition, because seasonal stochastic dependence and known calendar shocks can alter the forecast path. The gap addressed in this project is therefore not the invention of a novel forecasting algorithm. It is the disciplined implementation of a store-level forecasting design that aligns three things simultaneously: the operational decision unit, the timing of usable information, and a fair model ladder rooted in weekly seasonality.

2.2 Core time-series concepts underlying the project

Because this is a time-series forecasting problem, several theoretical concepts matter directly for the empirical design. The first is weak stationarity. A weakly stationary process has a constant mean, a time-invariant variance, and autocovariances that depend only on the lag between observations rather than on calendar time itself. In practice, restaurant demand rarely appears stationary in raw levels. Daily visitors can differ across weekdays, local demand conditions can drift over time, and some stores exhibit sustained shifts in average traffic. A model that ignores these features risks confusing persistent structure with unpredictable noise.

The second concept is autocorrelation. Let $\{z_t\}$ denote the transformed visitor series for one store. The lag-k autocorrelation function is

$$\rho_k = \frac{\mathrm{Cov}(z_t, z_{t-k})}{\mathrm{Var}(z_t)}.$$

If weekly routines matter, then ρ_7 should be economically important because the same weekday one week earlier contains information about the current day. Seasonal Naive is built directly on this intuition. However, the presence of autocorrelation at lag 7 does not imply that the correct model is simply to copy the past value. It only implies that dependence exists. SARIMA takes the next theoretical step by treating that dependence as part of a stochastic process that can be modelled through seasonal differencing and seasonal ARMA terms.

The third concept is differencing. In time-series analysis, differencing is used not because it is fashionable, but because a forecasting model for a non-stationary series can otherwise generate misleading dependence estimates and unstable forecasts. Non-seasonal differencing,

$$\nabla z_t = (1 - B)z_t = z_t - z_{t-1},$$

removes local level shifts. Seasonal differencing at period 7,

$$\nabla_7 z_t = (1 - B^7)z_t = z_t - z_{t-7},$$

removes recurring weekly level effects. When both operators are used,

$$\nabla\nabla_7 z_t = (1 - B)(1 - B^7)z_t,$$

the model is effectively built on changes relative to yesterday and to the same weekday one week earlier. This is especially appropriate when the raw process exhibits both slow-moving level variation and strong weekly seasonality.

The fourth concept is the distinction between deterministic and stochastic seasonality. A deterministic seasonal view would treat weekdays as fixed mean shifts and little more. A stochastic seasonal view allows the same weekday effect to vary through time through autoregressive and moving-average propagation. In restaurant data, that second view is often more realistic. Fridays are not merely "high" and Mondays merely "low" in a fixed mechanical sense. Rather, each weekday pattern interacts with recent demand shocks, local momentum, and residual dependence. This is one reason why SARIMA can outperform a purely deterministic weekly copying rule.

The fifth concept is forecast-origin discipline. A forecast is meaningful only with respect to an information set available at the time of prediction. Formally, if $\mathcal{F}_t$ denotes the information available at date t, then the forecasting problem is to form $E(y_{t+h} \mid \mathcal{F}_t)$ for horizon h. This principle explains both the chronological splitting strategy and the restriction to forecast-origin-valid exogenous variables. The public-holiday indicator satisfies this condition exactly because the calendar is known in advance. In contrast, a variable that becomes visible only after the forecast date would not belong in the information set and would invalidate the forecasting exercise.

These concepts directly motivate the final comparison. Seasonal Naive uses weekly autocorrelation mechanically. SARIMA models that weekly process through differencing and stochastic dependence. Holiday-only SARIMAX then asks whether, conditional on that stochastic structure, a known calendar intervention shifts the forecast further. The report is therefore best understood as an application of core time-series principles rather than as a generic predictive exercise.

2.3 Theoretical framework: demand under uncertainty

The theoretical framework treats restaurant visitor forecasting as a decision problem under uncertainty rather than as a purely statistical exercise. Managers choose labour, ingredients, and table capacity before realised demand is known, so forecasting matters because it reduces mismatch between expected arrivals and prepared capacity.

Within that decision problem, three channels are especially important. The first is store-specific persistence. A restaurant that has recently attracted many customers may continue to do so because of local reputation, cuisine type, location, or habitual clientele. The second is calendar structure. Workdays and public holidays change opportunity costs and leisure patterns, so demand is not equally likely on all dates. The third is the broader forecasting information set: some external variables may matter only if they are known before the forecasted day and only if they explain variation not already contained in the store's own history.

These channels map directly onto the empirical comparison in the project. Seasonal Naive lag 7 captures the simplest possible version of weekly structure by repeating the same weekday from the previous week. SARIMA captures that weekly structure in a stochastic time-series form through seasonal differencing and seasonal ARMA terms. Holiday-only SARIMAX then asks whether one known calendar shock, `is_holiday`, improves prediction further once internal seasonal dependence is already modelled. The model comparison therefore has a clear theoretical interpretation: if SARIMA outperforms the weekly benchmark, seasonal dependence matters beyond literal copying; if holiday-only SARIMAX improves further, a known calendar signal adds predictive value beyond that internal dynamics.

The framework also constrains the type of claim the report is allowed to make. This is a forecasting design, not a causal identification design. Holiday indicators are included because they are known before the forecasted visit date and therefore belong in the forecast information set. The question is whether they improve prediction under realistic timing, not whether the report identifies their causal effect on visitor demand. This distinction is not cosmetic. It is what keeps the theory, variables, and empirical claim aligned.

The same framework can also be written in conditional-expectation form. Let $y_{i,t}$ denote visitors for store i on day t, and let $\mathcal{F}_{i,t-1}$ denote the information set available for that store at the end of day $t-1$. The forecasting problem is to approximate

$$E(y_{i,t} \mid \mathcal{F}_{i,t-1}).$$

Seasonal Naive approximates this expectation with a highly restricted rule: yesterday's information matters only through $y_{i,t-7}$. SARIMA enlarges the information extracted from the store's own history by allowing a filtered combination of lagged values and lagged shocks after differencing. Holiday-only SARIMAX enlarges the information set one step further by adding a deterministic, known calendar variable. This progression clarifies the logic of the ladder: each model expands the approximation to the conditional expectation in one theoretically interpretable direction.

This matters because good time-series modelling is not only about lowering error; it is about matching the forecast structure to the information set. In this report, theory disciplines both which variables enter the model and the order in which modelling complexity is introduced.

2.4 Hypothesis derived from the framework

The theoretical framework implies one main hypothesis with a deliberately narrow scope: a store-level seasonal time-series model should forecast daily visitors more accurately than a Seasonal Naive lag-7 benchmark, and a parsimonious holiday regressor may improve further if public-holiday information contains predictive content beyond internal weekly dynamics. This is not a claim that SARIMAX is universally optimal or that every exogenous regressor should help. It is a more disciplined claim: within the implemented design, gains should first appear from modelling seasonal stochastic structure correctly, and any further gain from exogenous information should come only from variables that are both forecast-origin valid and economically meaningful.

That main hypothesis can be decomposed into three expectations that connect the theory to the empirical test:

- H1: weekly seasonality is strong enough that a lag-7 benchmark provides a meaningful forecasting standard;
- H2: a seasonal ARIMA specification should improve on the benchmark by modelling dependence beyond literal last-week repetition;
- H3: holiday information may provide predictive value beyond internal weekly dynamics.

Stating the hypothesis this way makes the benchmark part of the theory rather than just a technical comparator. It also clarifies that the report is not trying to settle which forecasting paradigm is best in general. It tests whether one coherent store-level design, motivated by the economics of restaurant demand, can improve on a strong weekly standard under chronological evaluation.

Table 1: Theoretical channels and empirical representation

Theoretical channel	Operational interpretation	Empirical representation
Store-specific persistence	Demand is partly shaped by recent store-level history	AR and MA terms in SARIMA / SARIMAX
Weekly calendar structure	Customer behaviour differs by weekday	Seasonal period = 7 and lag-7 benchmark
Holiday shocks	Public holidays can shift eating-out behaviour	`is_holiday`
Advance demand signal	Richer future information may matter in broader designs	Not used in the final compact model ladder

2.5 Testable implications of the framework

The framework implies several empirical patterns that make the later evaluation interpretable rather than arbitrary. First, the Seasonal Naive model should be a meaningful baseline, not a trivial one. If weekly seasonality is genuinely strong, then last week's same weekday should already provide a reasonable first approximation of future demand. Second, SARIMA should improve pooled RMSLE if the remaining dependence structure is better modelled as a stochastic seasonal process than as a pure copying rule. Third, a holiday-only SARIMAX specification should improve further only if public holidays shift demand in a way not already absorbed by the seasonal dynamics. A pooled gain without store-level consistency would be less persuasive because it could be driven by a narrow subset of high-volume or unusually predictable restaurants.

These expectations also explain why the report uses more than one summary statistic. Pooled RMSLE tests whether each successive model lowers average log-scale error across the operational prediction rows. The store-level win rate tests whether the improvement is broadly distributed across the panel rather than concentrated in a few cases. The rolling-origin folds test whether the same ordering reappears at multiple forecast origins rather than being an artefact of one holdout window. Together, these criteria connect the empirical evidence back to the theory of the problem.

2.6 From theory to empirical design

The empirical design follows directly from the theoretical structure of the problem. Persistence implies that recent store history must enter the model, weekly seasonality implies a seven-day cycle in both the benchmark and the final specification, and holiday effects imply that known calendar variables belong in the forecasting information set. Because the operational decision is made at the store-date level, both modelling and evaluation are also conducted at the store-date level.

The final pipeline is therefore not a collection of disconnected technical choices. Seasonal Naive is the null model for repeated weekly structure, SARIMA is the benchmark for internal seasonal dependence, and holiday-only SARIMAX is the parsimonious extension that adds one predetermined external signal.

Chronological evaluation is required because forecasting is defined by information timing, and pooled RMSLE is appropriate because the target is a non-negative, heterogeneous count process observed across many stores.

There is also a methodological reason to favour this ladder. A weekly benchmark may be too restrictive because it treats dependence as exact repetition, while a feature-rich model may blur the distinction between internal dynamics and external regressors. The ladder is valuable because it introduces complexity in interpretable steps; if the second step already captures most of the gain, that is itself a substantive result.

3. Data and Methodology

3.1 Data source and final sample

The empirical analysis uses the Recruit Restaurant Visitor Forecasting dataset (Kaggle, 2017), which combines restaurant visits, reservations, store characteristics, and calendar information. The report retains the Kaggle prediction logic at the store-date level: the target is daily visitors for each `air_store_id`, not one aggregate series for the entire market.

The full AIR panel contains 814 stores. After applying the modelling requirements, 812 stores remain eligible. The final report evaluates a main sample of 250 eligible stores. This sample is not treated as a provisional trial but as the report's primary empirical sample. In practice, the 250 stores are a fixed subset of eligible AIR stores carried through the final holdout and all three rolling-origin folds with the same model ladder and scoring rules. The purpose of that design is to preserve a coherent, computationally tractable store-level comparison while keeping runtime manageable and the resulting evidence directly reproducible from one consistent notebook pipeline.

The selection rule is deterministic and uses only pre-model information. In the final notebook, eligible stores are sorted by observed sample size, then by available pre-holdout training days, and then by store identifier; the first 250 are retained. That matters for interpretation because the sample is not chosen on forecast accuracy, residual behaviour, or any ex post measure of model success. It is therefore better understood as a stable, high-information subset of the eligible panel rather than as a hand-picked set of favourable stores.

This also clarifies what "representative" should mean in the context of the report. The 250-store sample is not claimed to be a simple random sample of all restaurants in Japan or even of all 812 eligible AIR stores. It is representative of the part of the panel most suitable for repeated store-level seasonal modelling under a common specification: stores with relatively long observed histories and dense enough data to sustain the same holdout and rolling-origin design. Numerically, the sample covers about 30.8% of eligible stores (250/812) and about 30.9% of eligible holdout observations (9,721/31,476). That share is large enough to make the final comparison evidentially meaningful, while the deterministic rule keeps the scope of the claim transparent.

A key construction choice is that each store series is built only over its observed date span rather than being padded with synthetic pre-history. Missing days inside that observed span are then filled as zero visitors in order to create the regular daily spacing required for both the lag-7 benchmark and the SARIMA-family specifications. This choice is a modelling convention, not a literal behavioural claim about every unobserved raw-data day. Its role is to define a regular daily state space in which lag 7, seasonal differencing, and seasonal ARMA dynamics are well defined. The global visit range in the final notebook runs from 1 January 2016 to 22 April 2017, but the effective modelling window differs by store because each series begins at its own first observed date. The final training period ends on 14 March 2017, and the final holdout runs from 15 March 2017 to 22 April 2017. On the 250-store main sample, the pooled holdout contains 9,721 observed store-date rows.

Table 2: Final sample definition

Item	Final project value
Forecasting unit	`air_store_id` × `visit_date`
Stores total	814
Stores eligible after modelling requirements	812
Stores in main evaluation sample	250
Global visit range	2016-01-01 to 2017-04-22
Final train end	2017-03-14
Final holdout window	2017-03-15 to 2017-04-22
Holdout observations	9,721 pooled observed store-date rows
Model fitting failures in main sample	0

3.2 Data cleaning and feature construction

Feature construction is organised around one central requirement: SARIMA-family models need a regular time index. The final notebook therefore builds a daily frame for each store over that store's observed span. The dependent variable is daily visitors, and the transformed modelling target is `log1p_visitors`. This transformation reduces skewness, handles zero counts naturally, and aligns the modelling scale with the report's preferred evaluation metric, RMSLE.

The broader project pipeline also constructed richer exogenous variables, including reservations and additional calendar indicators. Those variables remain useful for understanding the design space, but the final compact model ladder in this report uses only the holiday indicator as the exogenous regressor. This narrowing is itself informative from a time-series perspective. It reflects the empirical finding that the key comparison is not between one naive baseline and one feature-rich model, but between internal seasonal dynamics and a parsimonious external calendar signal.

The final notebook builds each store only within its observed date span and fills internal missing dates as zero visitors. This should be understood as an operational panel-construction rule rather than as a literal claim that every missing raw-data date corresponds to true zero demand or confirmed closure. The important time-series point is simply that regular spacing is required for lag-7 benchmarking, seasonal differencing, and seasonal ARMA estimation; without a regular daily index, the stochastic structure implied by the model would be ill-defined.

Table 3: Core variables used in the final comparison

Variable	Definition in the project	Theoretical reason
`visitors`	Daily observed visitors for each `air_store_id`	Operational demand to be forecast
`log1p_visitors`	log(1 + `visitors`)	Stabilises scale and matches RMSLE
`is_holiday`	Public-holiday indicator	Parsimonious calendar shock used in final SARIMAX
Additional engineered features	Weekend, Golden Week, and reservation variables in the broader pipeline	Useful for broader model exploration but not retained in the final compact ladder

3.3 Empirical diagnostic evidence in the implemented pipeline

The project is theory-led, but its modelling choices are not imposed in a vacuum. The implemented design is supported by direct empirical diagnostics produced during the data-understanding stage and by the final forecasting comparisons. The direct diagnostics are representative rather than exhaustive: they are designed to establish whether the core time-series features required by the final model ladder are visible in the data, not to estimate a separate stochastic model-selection routine for every store. These figures are aggregate diagnostics used to motivate the common weekly seasonal structure adopted in the final model ladder; they do not replace store-level diagnostics, and forecast estimation and evaluation remain store-level throughout.

The first diagnostic concerns weekly seasonality. Figure 1 reports average visitors by weekday from the observed AIR store-date panel. The within-week pattern is clear: weekend and late-week demand are materially stronger than early-week demand. That evidence matters because a lag-7 benchmark is only sensible if the same weekday one week earlier contains information about the current day. A weekly benchmark would be theoretically unmotivated if no recurrent weekday structure were visible in the first place.

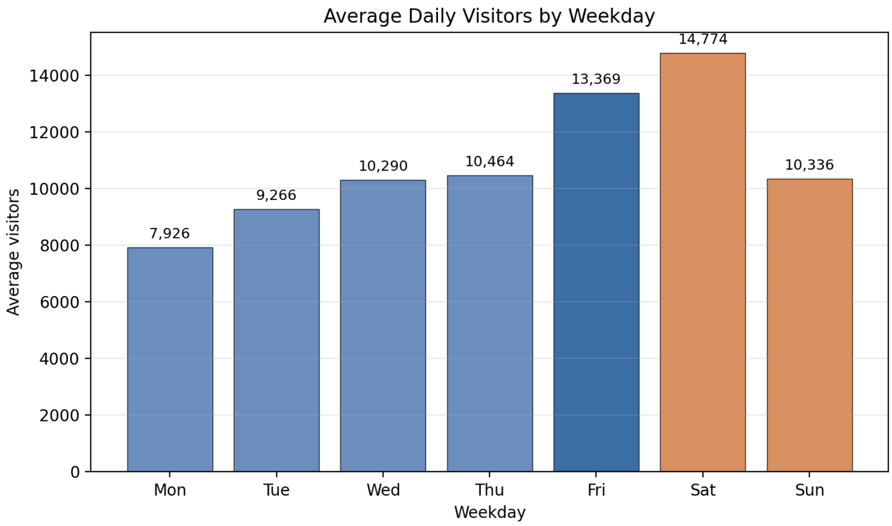

Figure 1: Average visitors by weekday across observed AIR store-date rows

The second diagnostic concerns serial dependence at the weekly frequency. Figure 2 reports the aggregate ACF and PACF of log(1 + `total daily visitors`) with lags marked at 7, 14, and 21 days. The correlogram shows slowly decaying dependence together with visible seasonal-lag spikes. In time-series terms, this is the empirical pattern one would expect when the process contains strong weekly persistence rather than isolated, one-period copying. This direct evidence supports two modelling choices at once: the use of seasonal period 7 and the move from a naive lag-7 rule to a stochastic seasonal model.

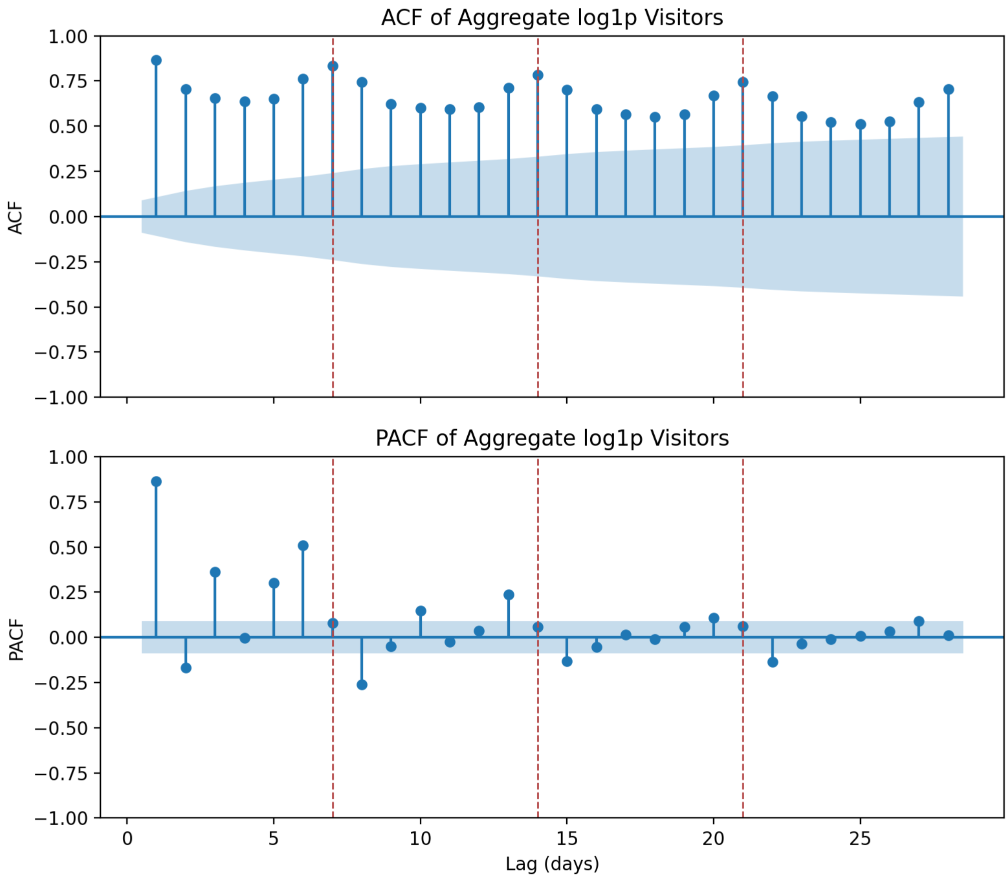

Figure 2: Aggregate ACF and PACF of $\log(1 + \texttt{total daily visitors})$

The third diagnostic concerns the need for seasonal filtering rather than undifferenced level modelling. Figure 3 gives the STL decomposition of the same aggregate log series with period 7. The seasonal component is not a negligible nuisance term; it is a prominent part of the observed process. Combined with the heterogeneity of store-level average visitor levels, this makes it plausible that raw-level modelling would confound local level shifts, weekly mean variation, and serial dependence. The use of both non-seasonal and seasonal differencing in SARIMA is therefore supported not only by textbook theory but by the observed behaviour of the data.

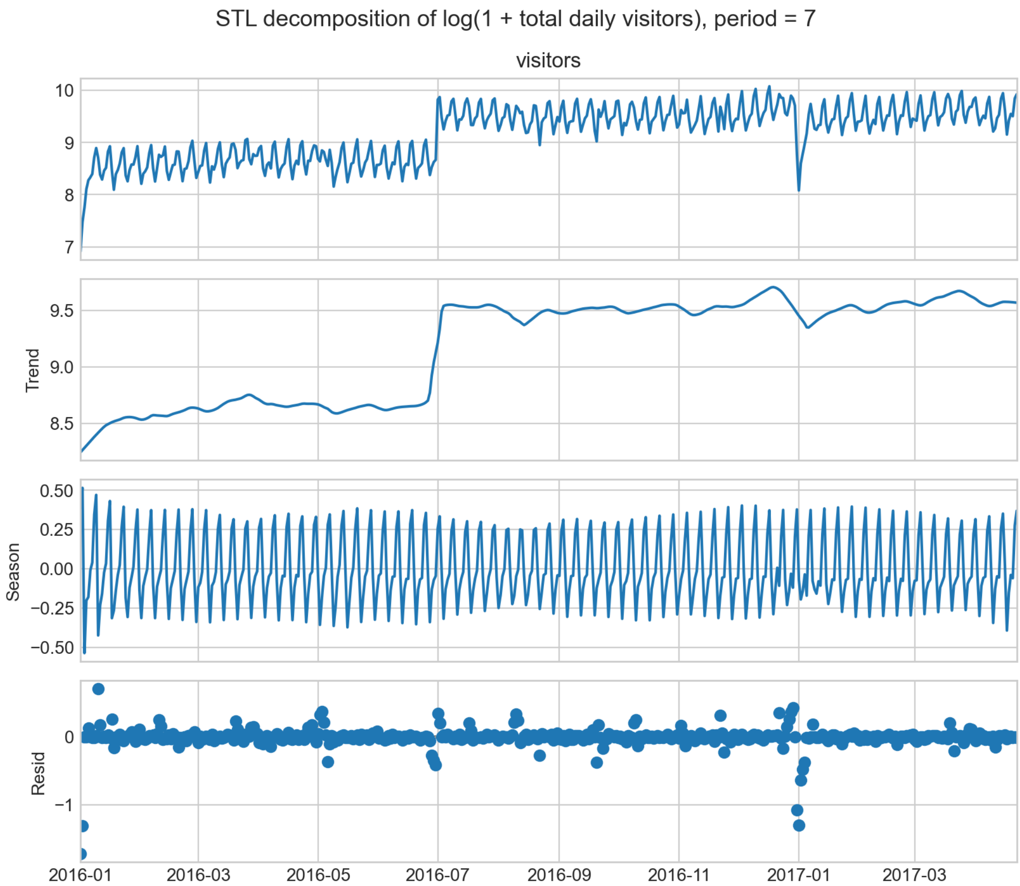

Figure 3: STL decomposition of $\log(1 + \texttt{total daily visitors})$ with seasonal period 7

The fourth diagnostic concerns what remains after the seasonal benchmark is fitted. Figure 4 reports the residual ACF from a representative aggregate SARIMA$(1, 1, 1)(1, 1, 1)[7]$ fit to the same log-transformed daily series. Relative to the raw-series correlogram, the remaining residual autocorrelation is materially weaker, which is consistent with the claim that the seasonal benchmark absorbs most of the weekly stochastic dependence. In time-series terms, this is the relevant residual-diagnostic standard for the benchmark used here: not literal perfection, but a marked attenuation of the dominant seasonal correlation that motivated the model in the first place. This diagnostic does not replace the forecasting comparison, but it does make the SARIMA benchmark more defensible as a genuine seasonal stochastic model rather than a purely mechanical alternative to the lag-7 rule.

Taken together, the implemented residual evidence is stricter than a single plot. First, the raw aggregate series shows the weekly dependence that the model is supposed to capture. Second, after seasonal differencing and SARIMA fitting, the residual correlogram is visibly flatter, so the benchmark is not leaving the original weekly pattern largely intact. Third, the out-of-sample results reinforce the same interpretation: SARIMA decisively beats Seasonal Naive on the final holdout and on every rolling-origin fold, which is what one would expect if the model is capturing genuine dependence rather than merely overfitting one estimation window. For a forecasting paper, that combination of pre-fit diagnostics, post-fit residual attenuation, and repeated out-of-sample superiority is a stronger residual argument than an isolated in-sample fit statistic alone.

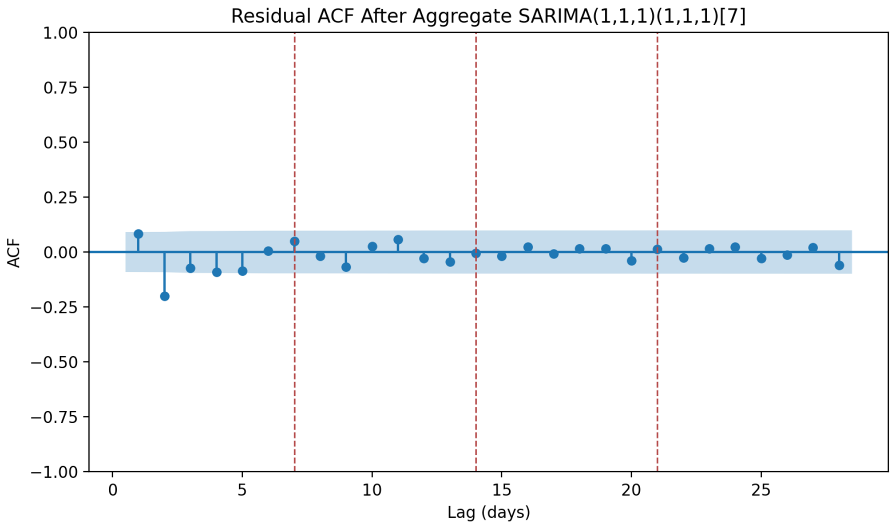

Figure 4: Residual ACF after aggregate SARIMA$(1, 1, 1)(1, 1, 1)[7]$ on $\log(1 + \texttt{total daily visitors})$

The fifth diagnostic concerns the exogenous signal. The holiday indicator is not introduced merely because holidays are intuitively important. It earns its place because, after the stochastic seasonal structure is modelled, holiday-only SARIMAX performs best on the final holdout and in two of the three rolling-origin folds. This is not a direct autocorrelation diagnostic, but it is a direct forecasting diagnostic of incremental information content: the holiday variable improves prediction after internal seasonal dynamics are already controlled for.

These diagnostic links between theory and evidence can be summarised compactly as follows.

Table 4: Empirical diagnostics supporting the final time-series design

Theory concept	Evidence in the implemented pipeline	Modelling implication
Weekly seasonality	Figure 1 shows systematic within-week visitor differences	Use lag-7 benchmark and seasonal period 7
Serial dependence at seasonal lags	Figure 2 shows persistence and visible spikes at multiples of 7	Model weekly structure as a stochastic seasonal process, not as literal copying alone
Seasonal signal in the observed process	Figure 3 shows a prominent weekly seasonal component	Support seasonal differencing and a seasonal SARIMA core
Residual seasonal dependence after benchmark fitting	Figure 4 shows weaker residual autocorrelation after the seasonal benchmark is fitted	Support the use of SARIMA as the main internal-dynamics benchmark
Forecast-origin-valid exogenous signal	Holiday-only SARIMAX is best on the holdout and in two of three folds	Retain `is_holiday` as a parsimonious external regressor

One limitation should still be stated openly. The diagnostic figures support the common weekly seasonal specification, but they do not provide a full store-by-store battery for all 250 stores. The report therefore does not claim universal residual whiteness across the panel. The residual claim is intentionally narrower: the implemented evidence shows that the benchmark meaningfully reduces dominant seasonal autocorrelation and performs robustly out of sample.

3.4 Operationalisation of theoretical channels

The theoretical framework becomes operational only when each channel is represented by variables with a clear forecasting role. The target variable, daily visitors, is realised restaurant demand at the store-date level. Because visitor counts are non-negative and right-skewed, the model uses the `log1p` transformation. This choice is not only convenient statistically; it also aligns the modelling scale with RMSLE, which measures error on a comparable log scale and reduces the dominance of very large stores.

Formally, if $y_{i,t}$ denotes observed visitors for store i on date t, the transformed target is

$$z_{i,t} = \log(1 + y_{i,t}).$$

Throughout the report, $y_{i,t}$ denotes visitor counts on the original scale, while $z_{i,t}$ denotes the log-transformed series used for estimation. This transformation preserves the non-negativity of the original count process while moving the estimation problem onto a scale that is less dominated by a small number of very large observations.

The transformation also changes the interpretation of forecast errors in a useful way. On the original count scale, an error of 20 visitors means something very different for a small restaurant than for a large one. On the transformed scale, differences are closer to proportional discrepancies. When the model mispredicts by a fixed amount in log space, the error corresponds roughly to a multiplicative mistake on the original scale. This is one reason why log-scale modelling and RMSLE are natural partners in a heterogeneous panel such as this one.

The holiday indicator corresponds to a specific calendar-demand mechanism. It represents known public-holiday disruptions to ordinary weekly patterns. In the final compact comparison, this variable is the only exogenous regressor carried into the seasonal dynamic-regression model, which the report refers to as holiday-only SARIMAX. That choice has a clear interpretation: the report is not testing whether a large feature set can be assembled, but whether one economically interpretable, forecast-origin-valid calendar variable improves a well-specified seasonal benchmark.

That choice also has a mathematical interpretation. In SARIMA, the model attributes predictable structure to internal lagged dependence after differencing. In SARIMAX, the exogenous regressor enters additively on the right-hand side and therefore shifts the conditional mean of the transformed process. In other words, the holiday indicator is not being asked to capture autocorrelation; it is being asked to explain a systematic calendar deviation after autocorrelation has already been modelled by the SARIMA terms. This separation of roles is good time-series practice because it avoids conflating serial dependence with exogenous level shifts.

Table 5: Operationalisation of theoretical channels in the final project

Theoretical channel	Operational meaning	Empirical representation	Expected forecasting role
Store-specific persistence	Recent store demand contains information about future demand	AR and MA terms in SARIMA / SARIMAX	Captures dynamic dependence within each store
Weekly seasonality	Demand differs systematically across weekdays	Seasonal period 7 and Seasonal Naive lag 7	Captures repeated weekly rhythm
Calendar structure	Public holidays can alter normal weekly demand	`is_holiday`	Allows one known date effect to shift forecasts
Advance demand revelation	Richer future information could improve forecasts in broader designs	Not included in the final compact model ladder	Outside the final report's main empirical claim
Scale heterogeneity	Restaurants differ greatly in visitor volume	`log1p_visitors`, RMSLE	Makes errors more comparable across stores

3.5 Model 1: Seasonal Naive lag-7 baseline

The baseline model is Seasonal Naive with lag 7. For each store, the forecast for a date is based on the observed visitor count from the same weekday in the previous week. This benchmark is deliberately strong rather than merely convenient. It directly captures weekly seasonality, which is one of the most persistent and economically plausible regularities in restaurant demand. Its strength is transparency and low specification risk; its limitation is that it assumes last week's pattern is sufficient even when the true data-generating process contains additional seasonal dependence.

In formal terms, the Seasonal Naive benchmark for store i can be written as

$$\hat{y}_{i,t|t-1} = y_{i,t-7},$$

so the forecast for date t is simply the realised demand from the same weekday one week earlier. The benchmark is therefore defined on the original visitor-count scale and later evaluated through the same pooled RMSLE criterion as the final models. In theoretical terms, this acts as a null model in which the dominant systematic component of demand is repeated weekly structure.

3.6 Model 2: Store-level SARIMA benchmark

The second model is SARIMA$(1,1,1)(1,1,1)[7]$ with no exogenous regressors, estimated separately for each store. The non-seasonal terms model short-run store-specific dynamics, while the seasonal component uses a seven-day period to capture weekly repetition. This specification corresponds closely to the theoretical framework developed in Section 2: persistence is handled by ARIMA dynamics and weekly structure is handled by the seasonal component.

Using the backshift operator B, where

$$By_t = y_{t-1},$$

the seasonal ARIMA family can be written compactly as

$$\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D y_t = \Theta(B^s)\theta(B)\varepsilon_t,$$

where s is the seasonal period, $\phi(\cdot)$ and $\theta(\cdot)$ are the non-seasonal autoregressive and moving-average polynomials, $\Phi(\cdot)$ and $\Theta(\cdot)$ are their seasonal counterparts, and ε_t is the disturbance term. In the implemented model, $s = 7$, $d = 1$, and $D = 1$, so the final store-level benchmark is

$$\Phi(B^7)\phi(B)(1-B)(1-B^7)z_t = \Theta(B^7)\theta(B)\varepsilon_t,$$

where $z_t = \log(1 + y_t)$. The equation makes the time-series structure explicit: differencing acts on the transformed demand process, while autoregressive and moving-average terms capture internal dynamics. This model is the correct benchmark if the relevant predictive information is already largely contained in each store's own past.

The order $(1, 1, 1)(1, 1, 1)[7]$ is used as a theoretically motivated fixed specification. The differencing terms address non-stationary level and seasonal patterns, the seven-day seasonality follows the dominant weekly structure, and the low-order AR and MA terms provide a parsimonious representation that remains computationally feasible across many stores. The design is therefore justified by a balance of theory, runtime, stability, and interpretability rather than by exhaustive per-store tuning.

Each parameter block has a distinct time-series meaning. The non-seasonal autoregressive term allows current differenced demand to depend on the previous period's differenced demand. The non-seasonal moving-average term allows current behaviour to respond to the most recent forecast error. The seasonal autoregressive term propagates dependence across one-week intervals after seasonal differencing, while the seasonal moving-average term allows the effect of a shock to persist at the weekly frequency. In a restaurant setting, this is appealing because demand shocks may echo not only day to day, but also from one weekly cycle to the next.

The differencing operators themselves can be written as

$$(1-B)y_t = y_t - y_{t-1}$$

and

$$(1-B^7)y_t = y_t - y_{t-7}.$$

The first removes local level variation, while the second removes recurring weekly level effects. The use of both operators therefore reflects a formal commitment to handling both ordinary and seasonal non-stationarity before interpreting the remaining dependence structure.

One can see this more concretely by expanding the combined differencing operator:

$$(1-B)(1-B^7)z_t = z_t - z_{t-1} - z_{t-7} + z_{t-8}.$$

This expression shows that the filtered series compares the current observation simultaneously to yesterday and to the same weekday last week. In practical terms, the transformed and differenced process asks whether current restaurant demand is unusually high or low relative to both its immediate local history and its weekly seasonal history. That is exactly the kind of benchmarked stochastic process one would expect to be informative in daily restaurant forecasting.

3.7 Model 3: Holiday-only SARIMAX extension

The third model keeps the same stochastic structure but adds one exogenous regressor. To make the forecasting logic explicit, it is useful to write the specification as a regression with SARIMA errors:

$$z_t = \beta_0 + \beta_1 \,\texttt{is\_holiday}_t + u_t,$$

where the disturbance process u_t follows

$$\Phi(B^7)\phi(B)(1-B)(1-B^7)u_t = \Theta(B^7)\theta(B)\varepsilon_t.$$

This representation is theoretically cleaner than writing the exogenous term directly inside the differenced equation, because it separates two roles that should remain distinct in time-series analysis. The holiday indicator shifts the conditional mean, while the SARIMA structure models autocorrelated dependence in the remaining disturbances. Because the disturbance process is seasonal, the more precise label for this specification is SARIMAX rather than a generic ARIMAX shorthand. The specification is deliberately compact. It asks a sharper question than a feature-rich design: once weekly seasonal dependence is already modelled by SARIMA, does a known holiday indicator still shift the forecast in a useful way? Because `is_holiday` is known before the visit date, the regressor satisfies forecast-origin discipline automatically. Its value is therefore interpretive as well as predictive: if it improves on SARIMA, the gain can be attributed to one clear calendar channel rather than to a diffuse bundle of exogenous features.

The coefficient β_1 has a forecasting interpretation rather than a causal one. On the transformed scale, it measures the average shift in the conditional mean associated with a holiday after seasonal differencing and stochastic dependence are already accounted for. If β_1 improves out-of-sample performance, the implication is not that the report has identified a structural holiday effect in a causal-econometric sense. The implication is narrower and more appropriate for forecasting: conditional on the information available at the forecast origin, holiday status contains predictive content not fully absorbed by the internal seasonal process.

Table 6: Implemented model comparison

Model	Role	Information used	Main limitation
Seasonal Naive lag 7	Weekly benchmark	Same weekday from previous week	Copies last week literally rather than modelling stochastic dependence
SARIMA$(1,1,1)(1,1,1)[7]$	Seasonal benchmark	Store history and weekly seasonal dynamics	Ignores known holiday shocks
Holiday-only SARIMAX$(1,1,1)(1,1,1)[7]$	Final compact model	SARIMA structure plus `is_holiday`	Adds only one external channel and does not claim to exhaust all possible exogenous information

3.8 Why the SARIMA-to-SARIMAX ladder is appropriate

A possible criticism is that different stores might prefer different ARIMA orders. That criticism is reasonable, but it does not invalidate the implemented design. The project chooses a fixed order as a trade-off between local optimality, comparability, runtime, and reproducibility. If every store had a separately tuned order, the project would introduce a second modelling layer: model selection for hundreds of individual stores. That could improve some store-level forecasts, but it would also increase complexity and overfitting risk.

The fixed order keeps the empirical question clean. The report is not asking whether the best possible ARIMA order can be found for every restaurant. It asks whether one coherent seasonal stochastic specification improves on a weekly copying rule, and whether one interpretable exogenous regressor improves further on that seasonal benchmark. The common order therefore makes the ladder meaningful: Naive isolates literal weekly repetition, SARIMA isolates internal seasonal dependence, and holiday-only SARIMAX isolates one external calendar channel.

The same logic applies to differencing. Store-level demand can shift over time, and weekly demand can exhibit recurring seasonal non-stationarity. Using both non-seasonal and seasonal differencing follows standard time-series logic: stabilise the stochastic process before interpreting its dynamic dependence. The move from SARIMA to SARIMAX then preserves the same stochastic core and asks whether a known holiday indicator shifts the conditional mean further.

This is why the ladder is more informative than a direct jump from Seasonal Naive to SARIMAX. Without the SARIMA step, any gain by the dynamic-regression model would be ambiguous. By inserting SARIMA explicitly, the report separates value from internal time-series structure from value contributed by one compact external calendar signal.

3.9 Forecast-origin discipline and missing-date treatment

A central requirement in time-series forecasting is that the model must not use information unavailable at the time the prediction is made. The final compact ladder is especially attractive in this respect because the only exogenous variable in the final SARIMAX specification is the public-holiday indicator. That variable is known in advance with certainty and therefore requires no special leakage treatment.

The treatment of missing dates raises a related methodological issue. Both SARIMA-family models and the lag-7 baseline require a regular daily time index. The final project avoids synthetic pre-history padding and models each store only over its observed date span. Within that span, missing days are filled as zero visitors. This is a pragmatic but defensible modelling assumption for three reasons: it preserves the regular index required by the benchmark and the seasonal state-space model; it avoids inventing pre-history before the first observed date; and, relative to interpolation or carry-forward rules, it is a conservative way to avoid creating artificial positive demand on unobserved internal dates. It should not be interpreted more strongly than that: the procedure creates a usable forecasting panel, but it does not by itself identify whether each missing raw record reflects closure, true zero demand, or incomplete observation.

3.10 Reproducibility

The empirical results are reproducible from the final modelling notebook and the processed inputs stored in the repository. The notebook reads from `notebooks/processed_data/`, writes outputs to `output/`, and stores the main-sample comparisons and fold summaries used in this report. The repository also includes a requirements file listing the core software dependencies. This organisation is intended to preserve evidentiary traceability by making clear which notebook, inputs, and outputs support the final claim.

The notebook also includes two optional extension blocks that are disabled by default: a deterministic 400-store robustness re-run using the same evaluation design, and representative residual diagnostics for selected stores using the fixed-order SARIMA benchmark. These extensions are part of the project infrastructure for further validation, but they are not counted as part of the default `main250cv` result set unless explicitly executed.

3.11 Evaluation strategy and metrics

Because the data are time series, evaluation is conducted through chronological splitting rather than random sampling. The primary reported metric is pooled RMSLE over all observed store-date rows in the final holdout. This metric is central because the forecasting task itself is defined at the store-date level and because the holdout contains many restaurants observed over the same final test window. Pooling all evaluated store-date errors therefore provides one overall measure of forecasting performance across the operational prediction units.

Formally, if y_j and $\hat{y}_j$ denote the observed and forecast visitor counts for evaluated store-date row j, pooled RMSLE is

$$\text{RMSLE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left[\log(1+\hat{y}_j) - \log(1+y_j)\right]^2}.$$

This definition shows explicitly why the metric aligns naturally with the transformed modelling scale used in the final specification. The models are estimated on the log-transformed series, while the reported evaluation is expressed as the root mean squared distance between realised and forecast counts on that same log scale.

The metric can also be motivated in decision-theoretic terms. In heterogeneous count data, absolute deviations on the raw scale overweight large-volume stores, while percentage measures become unstable near zero. RMSLE is a compromise that respects proportional errors more than absolute ones but remains well-defined when visitor counts are zero. Because restaurant demand in this dataset is both sparse for some stores and large for others, that compromise is analytically appropriate rather than merely convenient.

Chronological validation is necessary because the order of observations contains information. A random train–test split would be inappropriate: it could place later observations in the training set and earlier observations in the test set, creating an evaluation environment unlike the real forecasting problem. In practice, forecasts are always formed without knowledge of future realised demand. The final holdout therefore begins after the training period, and the rolling-origin folds preserve temporal order by construction. This follows the broader forecasting-evaluation logic emphasised by Tashman (2000), who argues that out-of-sample assessment should respect the temporal structure of the prediction problem.

From a time-series perspective, this is essential because forecast evaluation is fundamentally conditional on the information available at the origin date. A random split breaks that logic by allowing future states of the process to influence parameter estimation for earlier prediction targets. Such a design may still be useful in iid problems, but it is conceptually inappropriate here because it destroys the filtration structure that defines forecasting in the first place.

RMSLE is particularly suitable because visitor counts are non-negative and heterogeneous across stores. A raw error of 10 visitors has a very different meaning for a restaurant that usually serves 15 visitors than for one that typically serves 120. Log-scale errors reduce the dominance of large stores, accommodate zeros through the `log1p` transformation, and reflect the fact that proportional misses are often more meaningful than raw absolute deviations in operational forecasting.

MAPE is retained only as a secondary metric. It is intuitive because it is expressed as a percentage, but it becomes unstable when actual visitor counts are small. For that reason, the main judgement of the report is based on RMSLE and the store-level win rate rather than on percentage errors alone. Beyond the final holdout, the final notebook includes three rolling-origin folds: train through 14 January 2017 and test 15–28 January; train through 31 January and test 1–14 February; and train through 14 February and test 15–28 February. These folds are used to judge whether the model ordering is temporally stable rather than to replace the final holdout.

3.12 Methodological safeguards

The credibility of the project depends on four safeguards: chronological evaluation, a meaningful seasonal benchmark, a store-level forecasting unit, and a model ladder that separates internal seasonal dependence from external holiday information. Together, these safeguards make the final metrics interpretable and ensure that the supported conclusion remains aligned with the implemented design.

These safeguards matter because final metrics are persuasive only when the design that produced them is defensible. A strong benchmark prevents easy wins against weak comparators. Chronological evaluation prevents future information from contaminating the test period. The store-level forecasting unit keeps the empirical design aligned with the operational decision problem. The SARIMA-to-SARIMAX ladder clarifies whether gains come from stochastic seasonal structure or from exogenous information. Together, these choices reduce specification risk, information-set risk, and interpretation risk.

Table 7: End-to-end forecasting pipeline in the final project

Stage	Implemented action	Forecasting purpose
Raw data integration	Merge visitors, store metadata, and calendar variables into one store-date pipeline	Bring the core predetermined demand information into a regular forecasting frame
Panel construction	Build a daily frame within each store's observed span and fill internal missing dates as zero visitors	Create the regular spacing required for lag-7 forecasting and SARIMA-family estimation
Feature engineering	Construct `log1p_visitors` and `is_holiday` within a broader engineered pipeline	Operationalise the stochastic target scale and the compact calendar shock
Model estimation	Fit Seasonal Naive lag 7, SARIMA$(1,1,1)(1,1,1)[7]$, and holiday-only SARIMAX separately by store	Compare a weekly rule, a seasonal stochastic benchmark, and a parsimonious exogenous extension
Chronological evaluation	Score the final holdout and three rolling-origin folds using pooled RMSLE and store-level win rates	Keep the evidence aligned with realistic forecasting and store-level decisions

4. Results

4.1 Main pooled forecast accuracy

The primary empirical result is that both time-series models outperform the weekly benchmark on the 250-store main sample. On the final holdout, pooled RMSLE is 0.9314 for Seasonal Naive lag 7, 0.7255 for SARIMA no exog, and 0.7098 for holiday-only SARIMAX. Relative to the benchmark, SARIMA reduces RMSLE by 22.1%, while holiday-only SARIMAX reduces RMSLE by 23.8%. Relative to SARIMA, the holiday-only SARIMAX specification delivers a further 2.17% reduction.

This ordering is substantively meaningful because the comparison is layered. The move from Seasonal Naive to SARIMA shows that weekly restaurant demand is not adequately described by literal same-weekday copying alone. The smaller move from SARIMA to holiday-only SARIMAX then clarifies the role of exogenous information: most predictive gain comes from modelling seasonal stochastic structure correctly, while the holiday indicator provides a modest incremental adjustment when calendar conditions depart from ordinary weekly patterns.

Table 8: Main pooled forecast metrics on the final holdout

Model	RMSLE	RMSE_{log}	MAPE (%)
Seasonal Naive lag 7	0.9314	0.9314	128.40
SARIMA no exog	0.7255	0.7255	90.54
SARIMAX holiday only	0.7098	0.7098	87.49

4.2 Store-level consistency of the result

The pooled ranking is reinforced by the store-level evidence. On the final holdout, SARIMA beats Seasonal Naive for 87.2% of stores, holiday-only SARIMAX beats Seasonal Naive for 89.2% of stores, and holiday-only SARIMAX beats SARIMA for 67.2% of stores. These shares matter because the forecasting problem is operationally local. A restaurant manager cares less about one pooled metric in the abstract than about whether the model tends to improve store-level decisions across the panel.

The store-level comparison also sharpens the interpretation of the small RMSLE gap between SARIMA and holiday-only SARIMAX. The gain is not an artefact of a tiny number of extreme stores, because the holiday-only specification beats SARIMA for a clear majority of stores. At the same time, the win rate is not so overwhelming that one should describe the exogenous gain as dominant. The correct interpretation is therefore moderate: the holiday signal helps broadly, but it does not overturn the fact that the main forecasting work is already being done by the seasonal stochastic structure.

Table 9: Store-level summary of final holdout performance

Statistic	Value
Stores total	814
Eligible stores	812
Main evaluation sample	250
Holdout observations	9,721
SARIMA better than Seasonal Naive	87.2% of evaluated stores
Holiday-only SARIMAX better than Seasonal Naive	89.2% of evaluated stores
Holiday-only SARIMAX better than SARIMA	67.2% of evaluated stores

4.3 Rolling-origin validation

The report evaluates temporal stability using three rolling-origin folds on the same 250-store main sample. Fold 1 trains through 14 January 2017 and tests 15–28 January 2017. Fold 2 trains through 31 January 2017 and tests 1–14 February 2017. Fold 3 trains through 14 February 2017 and tests 15–28 February 2017. This design is materially stronger than relying on one supporting split because it asks whether the model ordering survives changes in forecast origin.

The fold results are supportive but not uniform. In Fold 1, RMSLE is 1.0777 for Seasonal Naive, 0.7345 for SARIMA, and 0.7162 for holiday-only SARIMAX, so the holiday-only specification is best. In Fold 2, the ordering reverses slightly at the top: RMSLE is 0.8679 for Seasonal Naive, 0.7020 for SARIMA, and 0.7115 for holiday-only SARIMAX, so SARIMA is best. In Fold 3, the two time-series models are nearly tied, with RMSLE 0.6802 for SARIMA and 0.6784 for holiday-only SARIMAX. Overall, the evidence is consistent with the holdout interpretation. Seasonal Naive is clearly weakest, SARIMA is a strong benchmark, and holiday-only SARIMAX is usually but not universally best.

This partial instability at the top of the ladder is not a weakness to hide; it is part of the time-series interpretation. Forecast performance varies by origin because the local state of the process varies by origin. If one validation window contains stronger holiday effects, the holiday regressor should help more. If another window is dominated by ordinary within-week dynamics, SARIMA may already be sufficient. The fact that the ranking shifts only slightly, while Seasonal Naive remains clearly weaker, is consistent with an exogenous gain that is useful but modest.

Table 10: Rolling-origin validation summary on the 250-store main sample

Fold	Test window	Seasonal Naive	SARIMA no exog	SARIMAX holiday only
CV1	2017-01-15 to 2017-01-28	1.0777	0.7345	0.7162
CV2	2017-02-01 to 2017-02-14	0.8679	0.7020	0.7115
CV3	2017-02-15 to 2017-02-28	0.9050	0.6802	0.6784

4.4 400-store holdout robustness check

Because the main report is based on a fixed 250-store evaluation sample, it is useful to ask whether the qualitative conclusion survives an expansion of the deterministic subset. The notebook therefore includes a larger holdout-only robustness run on 400 eligible stores selected by the same pre-model ordering rule. This extension keeps the final holdout window and the full model ladder unchanged, so the only substantive change is the breadth of the store panel.

The larger run preserves the core ranking. On the 400-store robustness sample, pooled RMSLE is 0.9209 for Seasonal Naive, 0.7178 for SARIMA no exog, and 0.7034 for holiday-only SARIMAX. Relative to Seasonal Naive, SARIMA improves by 22.06%, while holiday-only SARIMAX improves by 23.62%. Relative to SARIMA, the holiday-only gain is 2.00%. One model fallback is recorded under the forecast-sanitisation guardrail; this is a technical safeguard against numerically unstable forecasts and does not affect the main 250-store claim. Store-level win rates tell the same story: SARIMA beats Seasonal Naive for 88.0% of stores, holiday-only SARIMAX beats Seasonal Naive for 89.25% of stores, and holiday-only SARIMAX beats SARIMA for 65.5% of stores.

This robustness result sharpens the interpretation rather than weakening it. The dominant result remains stable: most forecasting value comes from moving from a weekly copying rule to a seasonal stochastic benchmark. The holiday-only specification still improves on SARIMA, and its incremental gain remains slightly smaller than in the 250-store main sample. The correct conclusion is therefore disciplined: the holiday channel is useful, but its contribution is still secondary relative to the more robust gain from the seasonal SARIMA core.

4.5 Representative residual diagnostics

The project also includes a small deterministic store-level residual check designed to complement the aggregate diagnostic figures. Six representative stores were selected from across the ordered eligible-store panel, and the same fixed-order SARIMA$(1,1,1)(1,1,1)[7]$ benchmark was re-fitted on each store's pre-holdout training sample. For each representative store, the notebook records residual ACF behaviour together with Ljung–Box p-values at lags 7, 14, and 21, which are the most relevant seasonal checkpoints for a weekly model.

The resulting pattern is supportive but not mechanically perfect. For five of the six representative stores, the Ljung–Box p-values at lags 7, 14, and 21 remain above the conventional 5% threshold, which is consistent with the view that the common seasonal benchmark removes most dominant weekly dependence in those cases. One representative store still shows statistically noticeable seasonal residual dependence at those lags. This is an informative result rather than a failure. It implies that the fixed-order benchmark is broadly appropriate for the implemented comparison, but it does not fully whiten every residual sequence at the store level.

This interpretation is consistent with the overall design of the paper. The common-order SARIMA benchmark is chosen to balance theory, comparability, runtime, and stability across many stores, not to optimise a separate model-selection problem for every restaurant. The representative residual diagnostics therefore strengthen the argument in the right way: they show that the benchmark is not simply winning on pooled forecast accuracy while leaving obvious weekly structure untouched in most representative cases.

4.6 Why the compact SARIMAX helps

The empirical pattern is exactly what one would expect from a well-constructed time-series ladder. Seasonal Naive captures only one source of predictability: weekly repetition. SARIMA improves sharply because it treats the series as a stochastic seasonal process rather than as a copying rule. Once that step is taken, there is less room for exogenous variables to contribute large gains. That is why the incremental improvement from holiday-only SARIMAX over SARIMA is modest rather than dramatic.

The holiday regressor helps because it captures a predictable deviation from ordinary weekday structure. In a seasonal ARIMA model, the dependence pattern is learned from past demand, but a public holiday is not merely another weekday repetition. It is a known intervention in the information set. In the language of time-series analysis, the exogenous regressor shifts the conditional mean when the calendar state changes, while the SARIMA terms continue to model the internal stochastic dependence. The fact that the gain is positive but small is itself informative. It implies that the weekly seasonal process already explains most of the variation that is predictable at the store level, and that public holidays refine rather than replace that structure.

There is also a parsimony lesson here. Every additional regressor must justify itself out of sample, not merely in narrative form. The compact SARIMAX specification performs well precisely because it adds one well-motivated variable without diluting the central role of the seasonal stochastic core.

4.7 What the results do not prove

The results must be interpreted within the boundaries of the implemented design. They do not prove that holiday-only SARIMAX is the best possible forecasting model across all alternatives, nor do they establish that the same ordering would necessarily hold on the full 812-store eligible panel or under every possible forecast origin. What they do show is more specific: on the main 250-store sample, seasonal stochastic modelling is clearly superior to a weekly copying rule, and a parsimonious holiday regressor usually improves further by a small amount.

4.8 Residual error interpretation and the role of MAPE

MAPE remains high even after the time-series improvements: 128.40% for Seasonal Naive, 90.54% for SARIMA, and 87.49% for holiday-only SARIMAX. Those values can look discouraging at first glance, but they must be interpreted in the context of store-level restaurant data, where some stores and days have very low visitor counts. When actual demand is close to zero, even a small absolute error can translate mechanically into a very large percentage error. That feature is intrinsic to MAPE rather than unique to this project.

RMSLE is more appropriate here because it better reflects the scale heterogeneity of the forecasting problem. It reduces the influence of very large stores, accommodates zeros through the `log1p` transformation, and matches the idea that proportionate misprediction often matters more than raw absolute deviation in operational settings. A miss of eight visitors does not have the same economic meaning across all restaurants, and RMSLE handles that asymmetry better than a simple percentage measure.

For that reason, MAPE is retained mainly as a secondary descriptive diagnostic. Its directional movement is still informative: both SARIMA and holiday-only SARIMAX improve MAPE in the same direction as RMSLE. But the substantive judgement of the paper rests on pooled RMSLE and store-level consistency, not on a metric known to become unstable in sparse count settings. This is a methodological choice grounded in the structure of the data rather than an ex post attempt to explain away an inconvenient result.

5. Discussion

5.1 Operational implications

Restaurant managers allocate labour, ingredients, and service capacity before realised demand is observed. In that setting, a forecast is valuable because it narrows the gap between expected and realised arrivals at the moment decisions must be made. The empirical results suggest that a model with well-specified seasonal time-series structure is much better aligned with that planning problem than a benchmark that only repeats last week's pattern.

This interpretation is strongest at the store level. Underprediction can lead to understaffing, slower service, and stock-outs, while overprediction can lead to excess labour cost and food waste. Those consequences materialise locally rather than in the aggregate. The incremental role of holiday information is also operationally sensible: public holidays are known in advance and can shift eating-out behaviour, but the evidence here suggests that they refine a strong seasonal core rather than replace it.

5.2 What this project shows about time-series modelling

Viewed as a time-series exercise, the project delivers three lessons. First, benchmark choice matters. A lag-7 benchmark already captures the most obvious weekly regularity, so beating it is non-trivial. Second, stochastic seasonal modelling matters more than raw feature accumulation. The movement from Seasonal Naive to SARIMA contributes the dominant improvement because it changes the way dependence is represented. Third, exogenous variables should be introduced parsimoniously and judged by incremental forecasting content, not by narrative appeal alone.

Pedagogically, the report reinforces the standard time-series sequence: identify the dependence structure, handle non-stationarity, respect the forecast origin, and only then ask whether an exogenous regressor adds incremental information.

5.3 Limitations and future extensions

The limitations of the study follow directly from its deliberately focused design. The first limitation is sample scope. The report evaluates a main sample of 250 eligible stores rather than the full 812-store eligible panel. That makes the comparison computationally tractable and internally coherent, but it also means the final numerical ranking should be interpreted as evidence for the selected evaluation sample rather than as a universal statement about every store in the dataset. The selection rule is deterministic and pre-model, which protects the design against performance-based cherry-picking, but the retained sample is still tilted toward stores with longer histories and larger observed spans. The 400-store holdout robustness check reduces this concern by showing that the qualitative ranking survives a materially larger subset, but it does not eliminate the underlying scope limitation. The report should therefore still be read as strongest for the high-information part of the panel rather than for the entire dataset without qualification.

The second limitation concerns temporal validation. Three rolling-origin folds are materially stronger than one auxiliary split, but they still do not establish performance across all possible forecast origins. In particular, the fact that SARIMA wins one fold while holiday-only SARIMAX wins two implies that the exogenous gain is supportive rather than absolute. Future work should extend the rolling-origin design further if the aim is to make stronger claims about stability.

The third limitation concerns panel construction. The notebook builds a complete daily frame within each store's observed span and fills missing internal dates as zero visitors in order to preserve the regular spacing required by the lag-7 benchmark and SARIMA-family models. This is a coherent operational rule, but it is still a modelling assumption rather than an economic observation. Future work could improve this part of the design by incorporating explicit store closure calendars, richer indicators of operating status, or alternative treatments of missing store-days.

The fourth limitation concerns model flexibility. The fixed order improves comparability, transparency, and reproducibility, but it may not be individually optimal for every store. Some restaurants may have dynamics that would be better captured by different orders or by models that borrow strength across similar stores. Future work could therefore compare the fixed-order design against automated per-store order selection, pooled panel models, or global forecasting systems.

Despite these limitations, the empirical design remains theoretically coherent. Demand is treated at the store level because the decision problem is store-specific. The benchmark uses lag 7 because weekly structure is central to restaurant demand. SARIMA models internal seasonal dynamics explicitly, while the holiday-only SARIMAX extension asks whether one forecast-origin-valid calendar regressor adds value on top of that structure. The outcome is transformed with `log1p` and evaluated primarily with pooled RMSLE because the forecasting task involves non-negative, skewed, heterogeneous count data.

6. Conclusion

This report has presented a store-level forecasting study of daily restaurant visitors using the Recruit Restaurant Visitor Forecasting dataset. The implemented comparison is intentionally focused: a weekly Seasonal Naive lag-7 benchmark, a store-level SARIMA$(1,1,1)(1,1,1)[7]$ benchmark with no exogenous regressors, and a holiday-only SARIMAX extension with the same stochastic core. The central question is not simply whether a more complex model wins, but where predictive gains come from once the benchmark itself is already strong.

The evidence supports a disciplined answer. On the 250-store main evaluation sample, most of the gain comes from moving from Seasonal Naive to SARIMA, which shows that restaurant demand is better modelled as a seasonal stochastic process than as literal last-week repetition. The holiday-only SARIMAX specification then improves modestly further, with pooled RMSLE falling from 0.7255 to 0.7098 on the final holdout and with the holiday-only model winning two of the three rolling-origin folds. The strongest defensible conclusion is therefore specific: in this store-level forecasting design, seasonal time-series structure does most of the predictive work, and public-holiday information provides a small but interpretable additional improvement.

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